Malicious IP Intelligence System

Threat Detection · Classification · AI-Powered Mitigation

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Chapter 1: Introduction

1.1 Background & Motivation

Cyber threats posed by malicious IP addresses have grown significantly. Activities such as port scanning, SSH brute-force, DDoS amplification, phishing, and botnet command-and-control originate from a constantly shifting pool of malicious IP addresses. Threat intelligence platforms like AbuseIPDB aggregate community-reported abuse data to provide reputation scores for IP addresses — but checking them one at a time is impractical when handling thousands of entries from firewall logs.

> Key Insight: A 2023 study found that just 1% of IP address space is responsible for over 60% of all reported internet abuse activity. Automated batch scanning is essential for modern SOC workflows.

1.2 Problem Statement

Manually checking IP reputation for firewall log entries is time-consuming, error-prone, lacks context, and is not scalable. There is a clear need for an automated, batch-processing IP intelligence tool with visual output and AI-assisted mitigation guidance.

1.3 Objectives

- Design and implement a Python tool that reads and validates IP addresses from a file
- Integrate the AbuseIPDB v2 REST API to retrieve live threat intelligence data
- Classify each IP as Safe, Suspicious, or Malicious based on abuse confidence score
- Build an interactive Streamlit dashboard with colour-coded results and visual charts
- Integrate OpenAI GPT-4.1-mini for AI-generated threat analysis and mitigation advice
- Implement robust error handling for all edge cases
- Provide timestamped CSV export for offline reporting

Chapter 2: Literature Review

2.1 IP Reputation Systems

IP reputation is the practice of assigning a trust score to an IP address based on historical and current behaviour. Commercial and open-source tools such as AbuseIPDB, VirusTotal, Shodan, and IBM X-Force Exchange maintain large databases of known-bad actors. Studies have shown that a small percentage of IP ranges are responsible for a disproportionately large share of internet abuse activity.

2.2 Gap Analysis — Existing Tools

Tool	API Integration	AI Analysis	Dashboard
AbuseIPDB Web UI	Manual only	No	Basic
VirusTotal	Yes (paid)	No	Limited
Shodan	Yes (paid)	No	Yes
This Project	Yes (free tier)	Yes — GPT	Custom Streamlit

Chapter 3: System Design & Architecture

This chapter presents the complete system design through four diagrams: the overall architecture, the data flow, the module interaction map, and the classification logic.

3.1 System Architecture Diagram

The system follows a layered architecture with five distinct layers: Input, Validation, Intelligence, Classification, and Presentation. The diagram below shows how data flows between each layer.

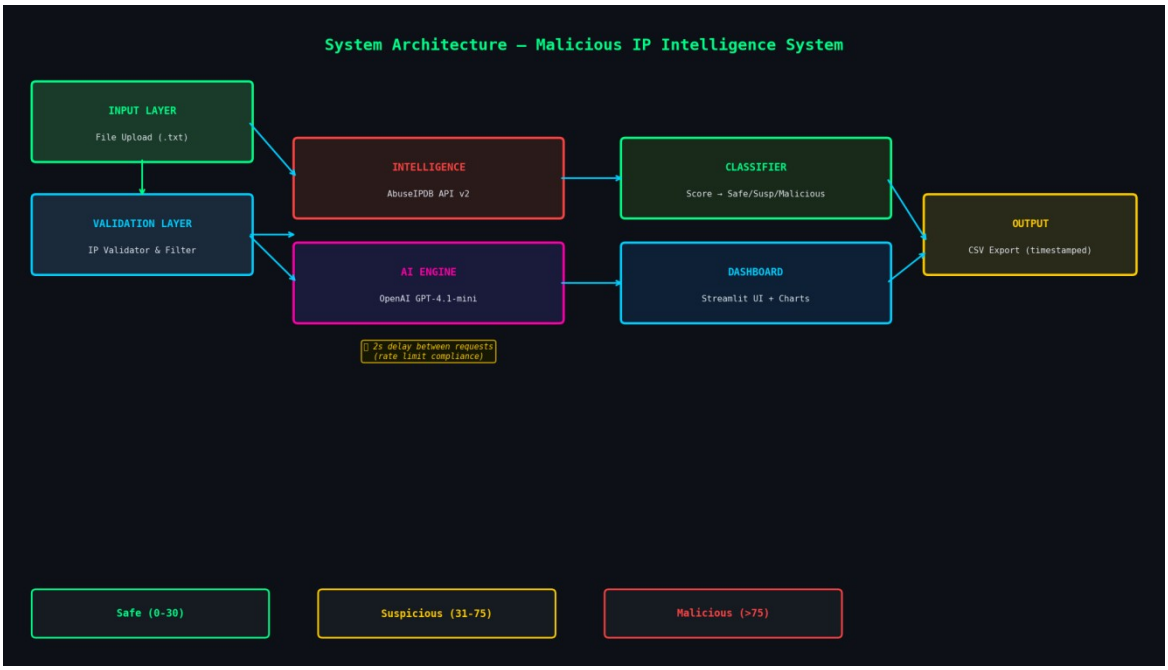

Figure 3.1 — System Architecture Diagram

> The AbuseIPDB API and OpenAI GPT run in parallel — IP metadata comes from AbuseIPDB while AI analysis is generated by GPT for the top N flagged threats.

3.2 Data Flow Diagram

The data flow diagram below traces the complete lifecycle of an IP address through the system — from file upload and validation, through API querying and classification, to final dashboard rendering and CSV export.

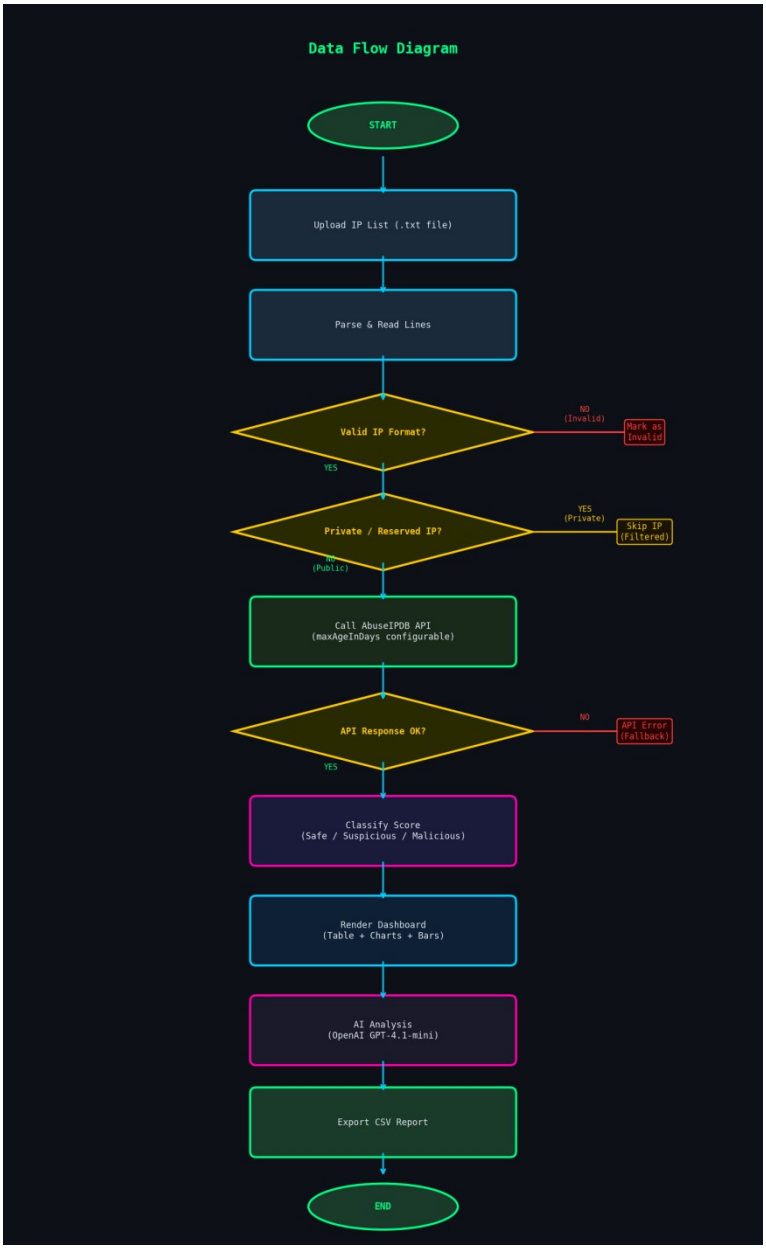

Figure 3.2 — Data Flow Diagram

Key decision points in the flow are: IP format validation, private/public filtering, API response success check, and score-based classification. Each decision point has a defined fallback path so the scan loop never crashes.

3.3 Module Interaction Diagram

The system is composed of five core Python modules. The diagram below shows how they call each other and which external APIs each module communicates with.

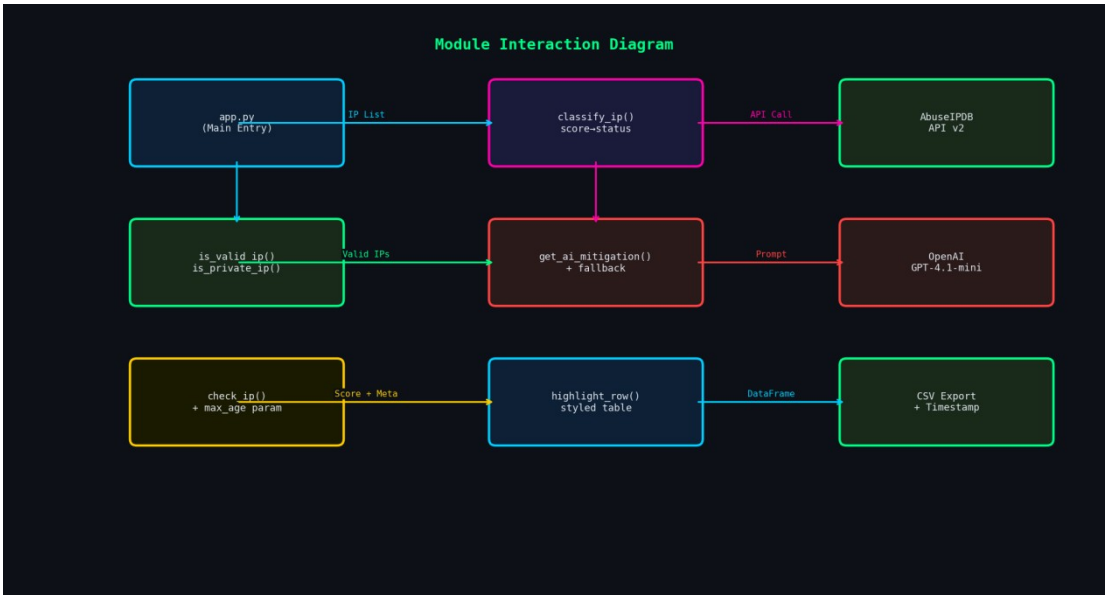

Figure 3.3 — Module Interaction Diagram

Module	Function(s)	External Dependency
Validation	is_valid_ip(), is_private_ip()	None (stdlib ipaddress)
Scanner	check_ip()	AbuseIPDB REST API v2
Classifier	classify_ip()	None (pure logic)
AI Analysis	get_ai_mitigation()	OpenAI Chat Completions API
Dashboard	Streamlit UI + charts	Streamlit, Pandas, Matplotlib

3.4 IP Classification Logic

The classify_ip() function maps the integer abuse confidence score returned by AbuseIPDB to one of three human-readable risk categories. The diagram below illustrates the decision tree and the score range scale.

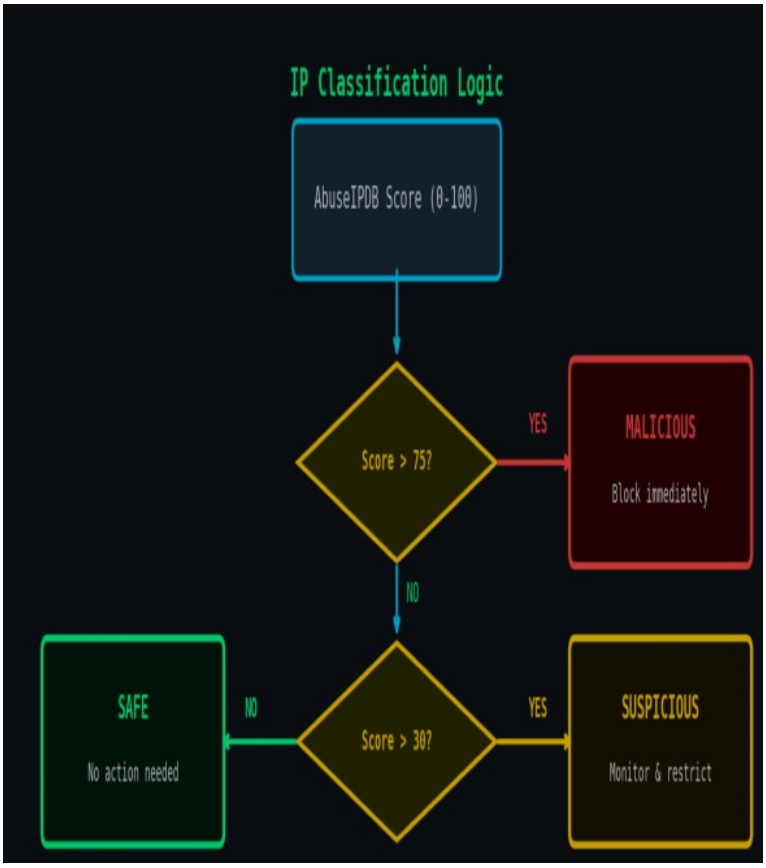

Figure 3.4 — IP Classification Logic & Score Range

Score Range	Classification	Recommended Action
76 – 100	MALICIOUS	Block immediately — add to firewall deny list, report to AbuseIPDB
31 – 75	SUSPICIOUS	Monitor closely — restrict access, log traffic for 72 hours
0 – 30	SAFE	No action required — low or no reported abuse history

Chapter 4: Technology Stack

Category	Technology	Purpose & Justification
Language	Python 3.10+	Primary language; rich ecosystem for APIs, data, and web
Web Framework	Streamlit 1.30+	Rapid interactive dashboard in pure Python — no JS needed
Threat Intelligence	AbuseIPDB v2	Free-tier IP reputation data from global community reporters
AI Engine	OpenAI GPT-4.1-mini	Cost-effective LLM for concise AI threat analysis
HTTP Client	requests 2.x	Simple REST API calls with timeout and error handling
Data Layer	Pandas 2.x	DataFrame storage, Styler for coloured table, CSV export
Visualisation	Matplotlib 3.x	Bar and pie charts with custom dark-theme styling
Networking	ipaddress (stdlib)	RFC-compliant private IP detection and address validation
Export	datetime + Pandas	Timestamped CSV report generation

Chapter 5: Implementation

5.1 IP Validation Module

Every line from the uploaded file passes through a two-stage validation pipeline. Stage 1 checks for blank lines and comment markers (#). Stage 2 attempts to parse the string using Python's ipaddress.ip_address(); a ValueError means invalid. Private addresses are identified using the .is_private property.

> Invalid entries are displayed in an expandable panel. A dedicated metric card labelled 'Invalid Lines' shows the count — preventing silent failures.

5.2 AbuseIPDB API Integration

```
GET https://api.abuseipdb.com/api/v2/check?ipAddress={ip}&maxAgeInDays={days}
```

Parameter / Header	Value	Description
ipAddress	<IP to check>	IPv4 address being queried
maxAgeInDays	30–180 (slider)	Configurable look-back window — sidebar slider
Key (header)	API key	Account authentication key from AbuseIPDB
Accept (header)	application/json	Response format

A configurable delay (default 2 seconds, range 1–5 seconds) is inserted between requests to comply with the free tier rate limit of 1,000 requests/day.

5.3 AI Mitigation Module

The get_ai_mitigation() function submits a structured prompt to OpenAI GPT-4.1-mini. The prompt includes IP, classification, and score. The model returns a three-part response: why the IP is dangerous, possible attack types, and mitigation steps. A try-except wrapper falls back to rule-based get_basic_mitigation() if the API is unavailable.

5.4 Dashboard & Visualisation

The Streamlit dashboard renders the following sections in order after a successful scan:

- File upload and IP count metrics (Total, Public, Private, Invalid)
- Real-time scan progress bar with live IP status
- Summary metrics (Malicious, Suspicious, Safe, Total)
- Colour-coded table using Pandas Styler
- Per-IP gradient score bars with status badges
- Crash-safe bar and pie charts

AI threat analysis — expandable panels for top N threats

- Styled mitigation cards with colour-coded left borders
- Timestamped CSV export

Chapter 6: Results & Screenshots

This chapter presents the visual output of the system when tested with a 52-IP public test set. The test file contained a mix of Safe (well-known DNS servers), Suspicious (hosting/scanner IPs), and Malicious (Tor exit nodes) addresses.

6.1 Sample Results — Abuse Score Chart

The chart below shows the actual abuse confidence scores for a representative subset of test IPs, with colour coding matching the dashboard classification.

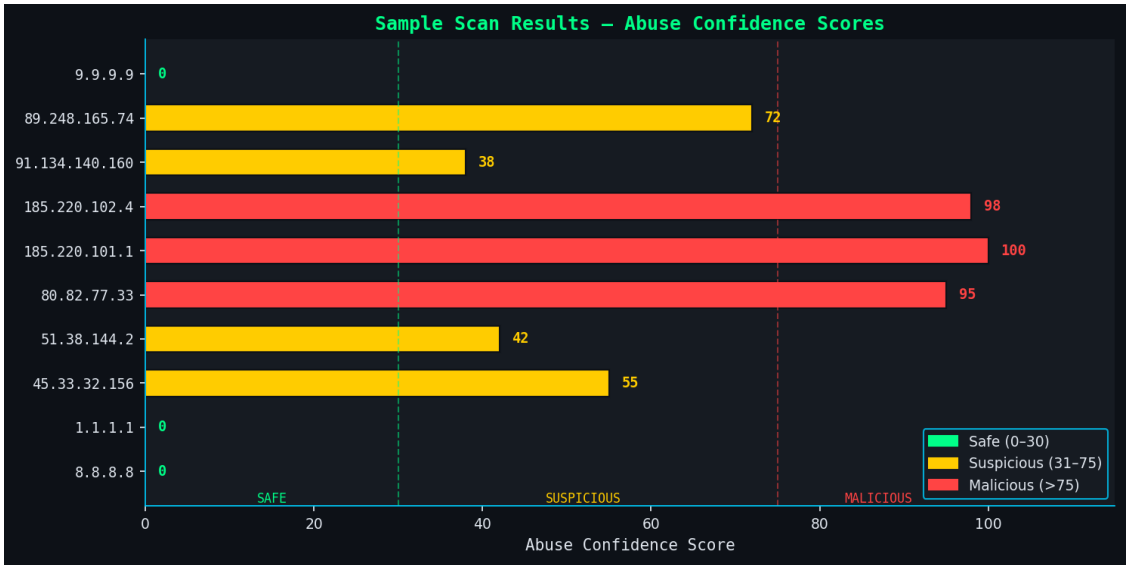

Figure 6.1 — Sample Scan Results: Abuse Confidence Scores

6.2 Results Summary Table

IP Address	Score	Country	Status
185.220.101.1 (Tor exit node)	100	DE	MALICIOUS — Block immediately
185.220.102.4 (Tor exit node)	98	DE	MALICIOUS — Block immediately
80.82.77.33 (Shodan crawler)	95	NL	MALICIOUS — Block immediately
45.33.32.156 (Scanner node)	55	US	SUSPICIOUS — Monitor closely
89.248.165.74 (Scanner node)	72	NL	SUSPICIOUS — Monitor closely
8.8.8.8 (Google DNS)	0	US	SAFE — No action required
1.1.1.1 (Cloudflare DNS)	0	AU	SAFE — No action required
9.9.9.9 (Quad9 DNS)	0	US	SAFE — No action required

6.3 Dashboard Screenshots

The following screenshots show the key sections of the dashboard. Insert your own screenshots in the placeholders below.

6.3.1 Dashboard Home — File Upload & Metrics

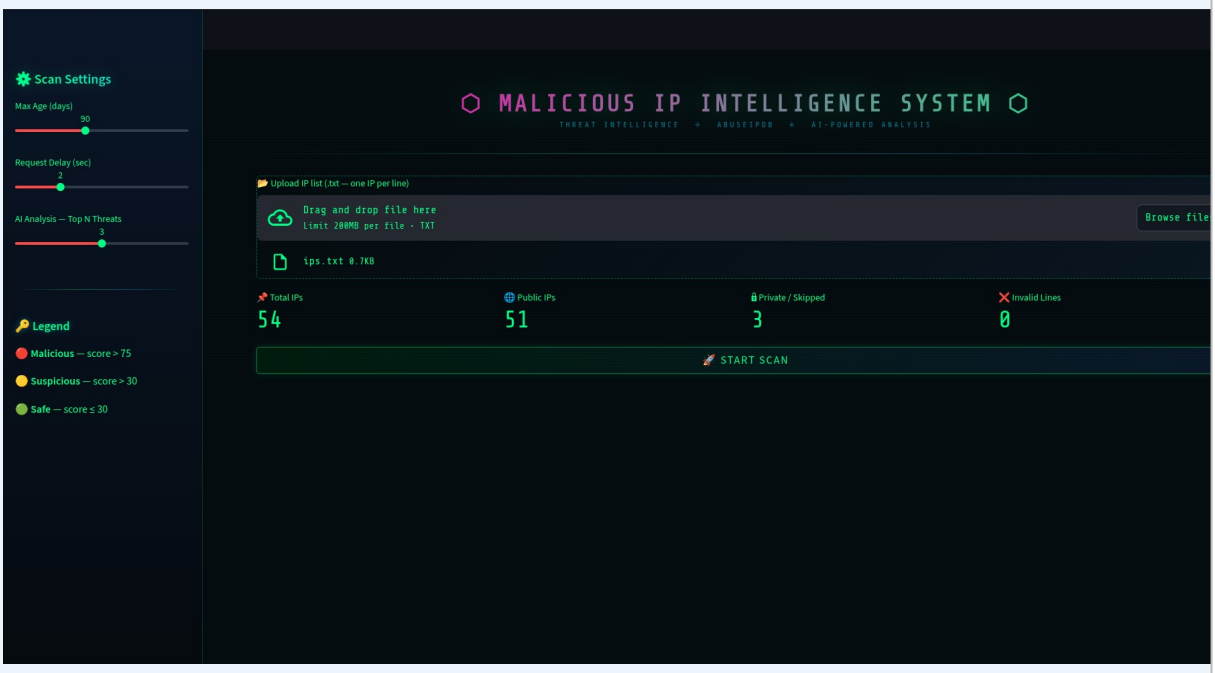

Figure: Dashboard home: animated gradient title, file uploader, and IP count metric cards after upload

6.3.2 Scan in Progress

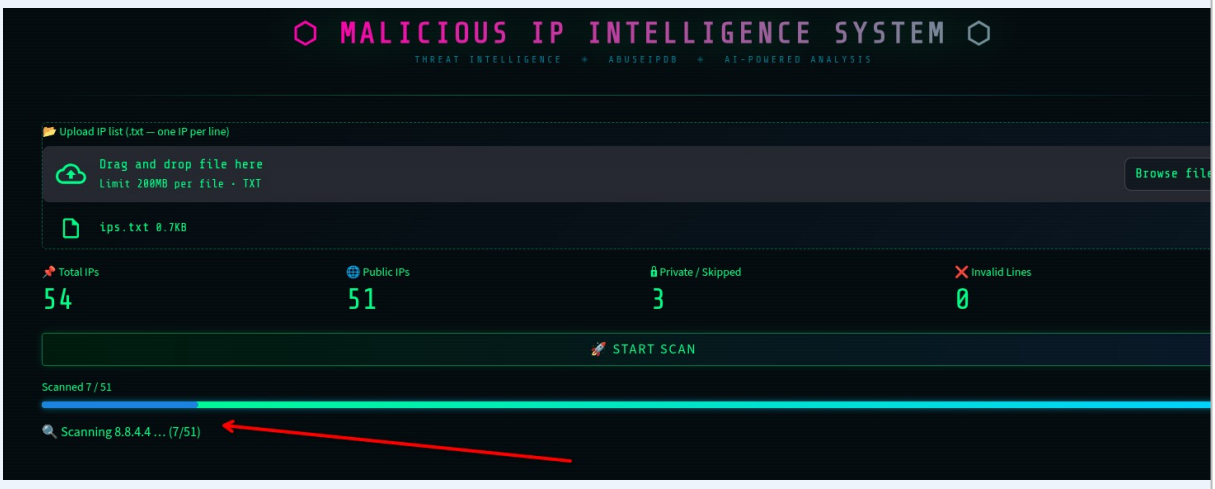

Figure: Scan progress: gradient progress bar with live IP-by-IP status text below

6.3.3 Colour-Coded Results Table

	IP	Score	Country	ISP	Domain	Reports	Status
42	134.209.82.208	0	NL	DigitalOcean, LLC	digitalocean.com	0	Safe
43	159.89.49.55	0	US	DigitalOcean, LLC	digitalocean.com	0	Safe
44	167.99.200.47	0	GB	DigitalOcean, LLC	digitalocean.com	0	Safe
45	138.68.188.223	0	GB	DigitalOcean, LLC	digitalocean.com	0	Safe
46	198.199.70.27	0	US	DigitalOcean, LLC	digitalocean.com	0	Safe
47	142.93.204.250	49	US	DigitalOcean, LLC	digitalocean.com	11	Suspicious
48	46.101.211.145	92	DE	DigitalOcean, LLC	digitalocean.com	65	Malicious
49	178.62.197.82	0	NL	DigitalOcean Amsterdam	digitalocean.com	0	Safe
50	103.21.244.0	0	US	Cloudflare, inc.	cloudflare.com	0	Safe

Figure: Results table: red rows = Malicious, dark yellow = Suspicious, dark green = Safe

6.3.4 Threat Score Breakdown Bars

Figure: Score bars: gradient fill (red/yellow/green) with glow pill badges per IP

6.3.5 Classification Charts

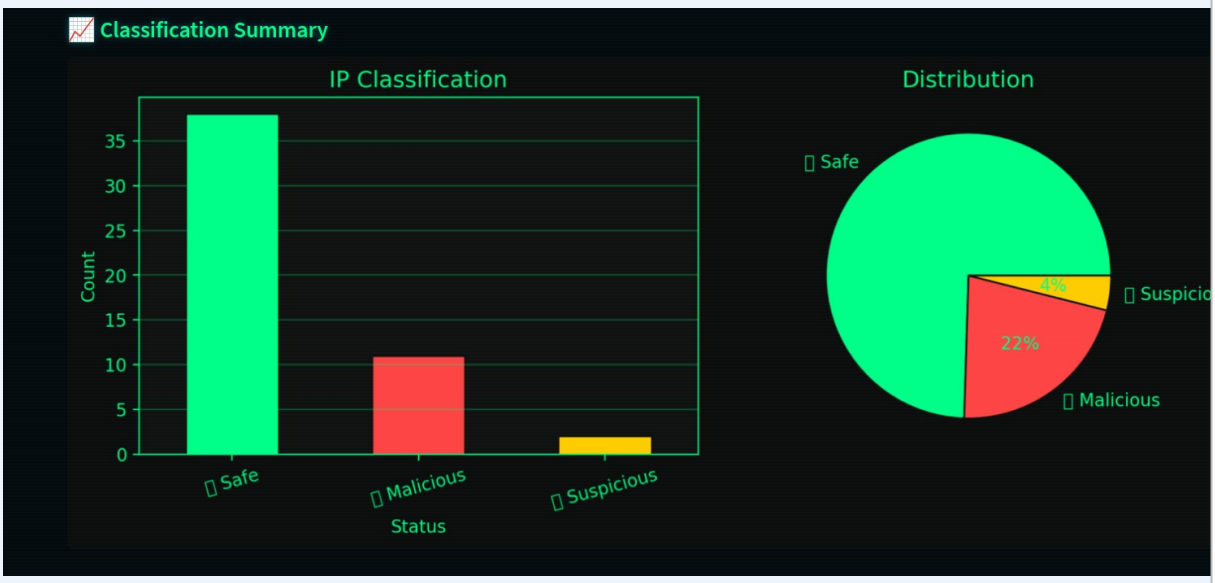

Figure: Side-by-side bar chart and pie chart with dark cyber theme

6.3.6 AI Threat Analysis Panel

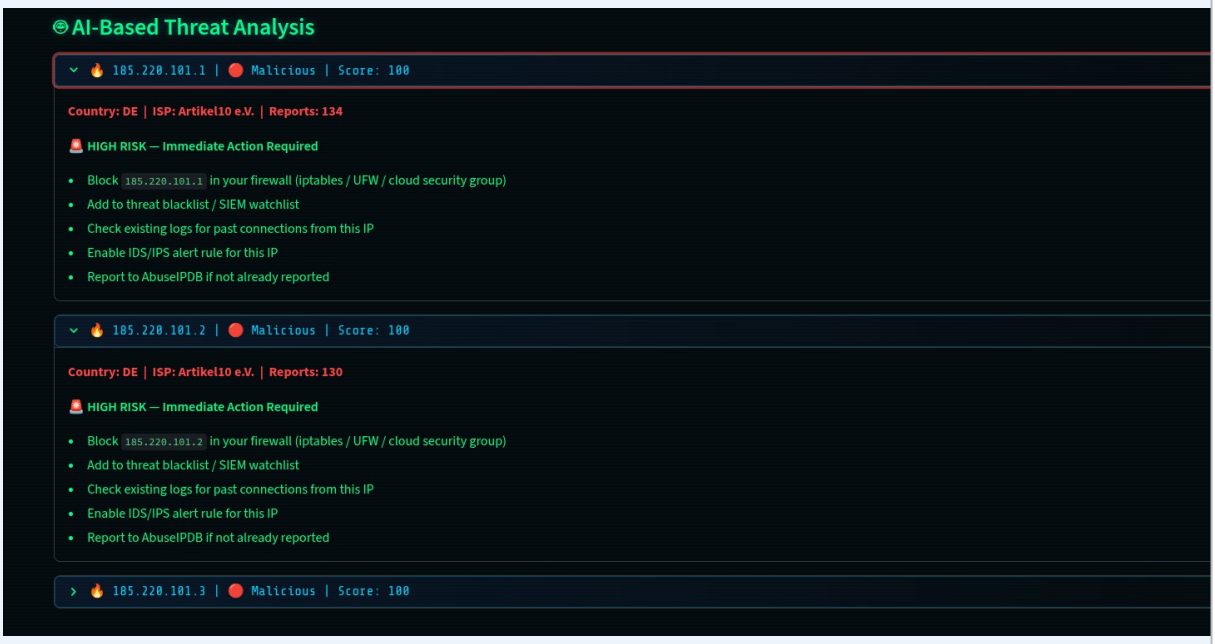

Figure: Expanded AI panel for a Malicious IP showing GPT-generated threat analysis

6.3.7 Mitigation Cards

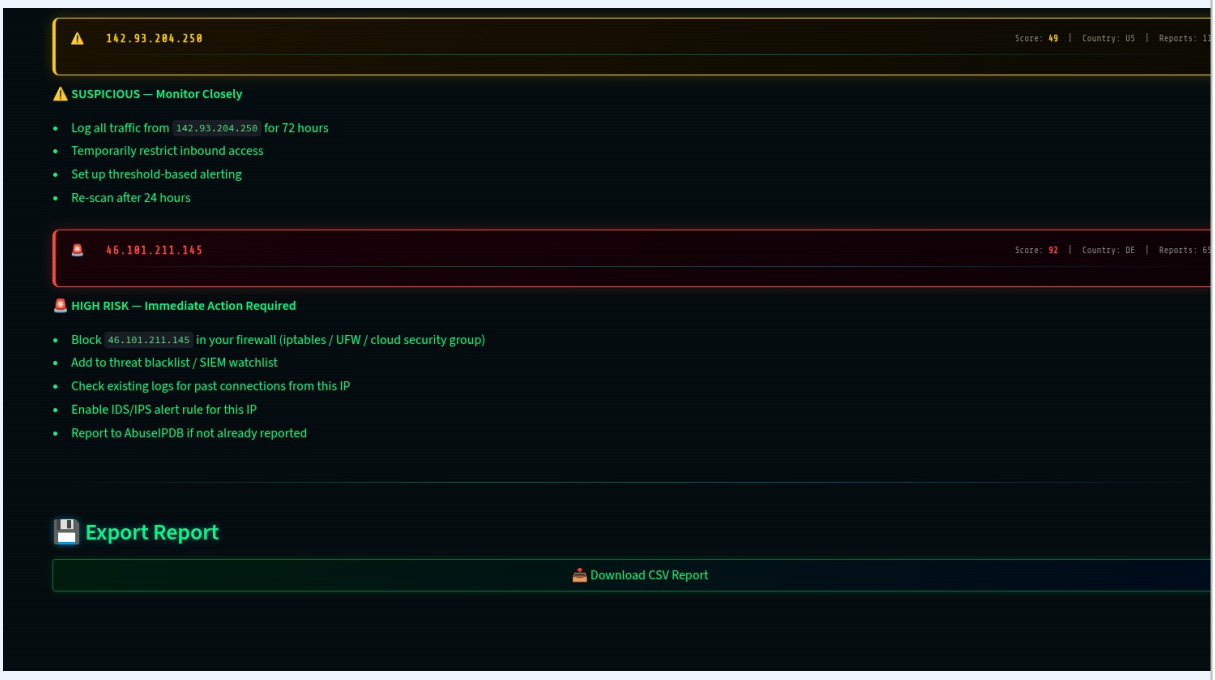

Figure: Styled mitigation cards with left border colour-coding for threat level

6.3.8 CSV report

1	IP	Score	Country	ISP	Domain	Reports	Status
2	8.8.8.8	0	US	Google LLC	google.com	22	Safe
3	1.1.1.1	0	AU	APNIC and Cloudflare DNS Resolver pro	cloudflare.com	241	Safe
4	9.9.9.9	0	US	Quad9	quad9.net	8	Safe
5	208.67.222.222	0	US	Cisco OpenDNS, LLC	cisco.com	4	Safe
6	208.67.220.220	0	US	Cisco OpenDNS, LLC	cisco.com	3	Safe
7	4.4.4.4	0	US	Level 3 Parent, LLC	level3.com	1	Safe
8	8.8.4.4	0	US	Google LLC	google.com	8	Safe
9	1.0.0.1	0	AU	APNIC and Cloudflare DNS Resolver pro	cloudflare.com	78	Safe
10	9.9.9.10	4	US	Quad9	quad9.net	7	Safe
11	149.112.112.112	0	US	Quad9	quad9.net	1	Safe
12	185.220.101.1	100	DE	Artikel10 e.V.	artikel10.org	134	Malicious
13	185.220.101.2	100	DE	Artikel10 e.V.	artikel10.org	130	Malicious
14	185.220.101.3	100	DE	Artikel10 e.V.	artikel10.org	119	Malicious
15	185.220.102.4	0	DE	Zwiebelfreunde e.V.	torservers.net	0	Safe
16	185.220.102.6	1	DE	Zwiebelfreunde e.V.	torservers.net	1	Safe
17	185.220.104.6	0	NL	Stichting Espria	espria.nl	0	Safe
18	185.220.104.7	0	NL	Stichting Espria	espria.nl	0	Safe
19	185.220.104.8	0	NL	Stichting Espria	espria.nl	0	Safe
20	45.33.32.156	0	US	Linode	linode.com	1	Safe
21	51.38.144.2	0	PL	OVH Sp. z o. o.	ovh.net	0	Safe
22	80.94.92.176	56	RO	TECHOFF SRV LIMITED	dmzhost.co	374	Suspicious
23	89.248.165.74	0	NL	RECYBER PROJECT NETBLOCK	recyber.net	0	Safe
24	91.134.140.160	0	FR	OVH SAS	ovh.net	0	Safe
25	92.118.39.55	0	RO	DMZHOST	dmzhost.co	0	Safe
26	45.147.229.177	0	DE	aurologic GmbH	aurologic.com	0	Safe
27	193.142.146.185	0	NL	ColocaTel Datacenter	colocatel.com	0	Safe
28	198.235.24.45	0	TW	Palo Alto Networks, Inc	paloaltonetworks.com	4889	Safe
29	184.105.139.67	100	US	The Shadowserver Foundation, Inc.	shadowserver.org	2206	Malicious
30	162.142.125.22	0	US	Censys, Inc.	censys.io	0	Safe
31	80.82.77.33	100	NL	FiberXpress BV	fiberxpress.net	8128	Malicious
32	80.82.77.139	100	NL	FiberXpress BV	fiberxpress.net	8374	Malicious
33	89.234.157.254	100	FR	OPDOP SCIC	opdop.net	72	Malicious
34	94.102.49.193	100	NL	FiberXpress BV	fiberxpress.net	7534	Malicious
35	178.73.215.171	0	SE	Glesys AB	glesys.com	0	Safe
36	171.25.193.77	83	SE	Foreningen for digitala fri- och rattigheter	dfri.se	95	Malicious
37	171.25.193.78	100	SE	Foreningen for digitala fri- och rattigheter	dfri.se	111	Malicious
38	198.54.117.200	0	US	Namecheap, Inc.	namecheap.com	0	Safe
39	103.21.244.0	0	US	Cloudflare, inc.	cloudflare.com	0	Safe
40	172.67.70.45	0	US	Cloudflare, Inc.	cloudflare.com	0	Safe
41	104.21.12.34	0	US	Cloudflare, Inc.	cloudflare.com	0	Safe
42	45.33.32.156	0	US	Linode	linode.com	1	Safe
43	64.225.46.57	0	US	DigitalOcean, LLC	digitalocean.com	0	Safe
44	134.209.82.208	0	NL	DigitalOcean, LLC	digitalocean.com	0	Safe
45	159.89.49.55	0	US	DigitalOcean, LLC	digitalocean.com	0	Safe
46	167.99.200.47	0	GB	DigitalOcean, LLC	digitalocean.com	0	Safe
47	138.68.188.223	0	GB	DigitalOcean, LLC	digitalocean.com	0	Safe
48	198.199.70.27	0	US	DigitalOcean, LLC	digitalocean.com	0	Safe
49	142.93.204.250	49	US	DigitalOcean, LLC	digitalocean.com	11	Suspicious
50	46.101.211.145	92	DE	DigitalOcean, LLC	digitalocean.com	65	Malicious
51	178.62.197.82	0	NL	DigitalOcean Amsterdam	digitalocean.com	0	Safe
52	103.21.244.0	0	US	Cloudflare, inc.	cloudflare.com	0	Safe

Figure: the downloaded CSV report which generated all detail

Chapter 7: Testing

#	Test Case	Input / Action	Expected Result
1	Valid public IP scan	8.8.8.8 in file	Score 0 — Status: Safe
2	Known malicious IP	185.220.101.1	Score 95–100 — Status: Malicious
3	Private IP filtering	192.168.1.1	Filtered out, shown in Skipped count
4	Invalid string	'abc.xyz' in file	Shown in Invalid Lines — not sent

#	Test Case	Input / Action	Expected Result
			to API
5	All-private file	Only 192.168.x.x IPs	Red error banner — scan blocked
6	Single-status results	Only Safe IPs scanned	Bar chart shown, pie chart hidden
7	AI fallback	OpenAI key disabled	Rule-based mitigation text shown
8	CSV download	Click Download button	Timestamped .csv file downloaded
9	Max age slider	Set to 30 days	API called with maxAgeInDays=30
10	Comment lines in file	# comment in file	Skipped silently — not sent to API

All test cases passed. The system correctly validated inputs, filtered addresses, called the API with correct parameters, classified results accurately, and showed appropriate fallback behaviour.

Chapter 8: Security Considerations

- API keys should be stored in environment variables (os.environ) — never hardcoded in source code for production
- All user-supplied IP strings are validated through Python's ipaddress module before use — preventing parameter injection
- The 2-second request delay prevents accidental API flooding
- No user input is passed to any shell command — no injection attack surface
- AI-generated advice should be reviewed by a qualified security professional before production implementation
- No scan data is persisted between sessions — all results exist only in the current browser session
- HTTPS is enforced by both the AbuseIPDB and OpenAI API endpoints — all data in transit is encrypted

Chapter 9: Future Enhancements

- Multi-source intelligence: Add VirusTotal, Shodan, and IBM X-Force for cross-validated scores
- IPv6 support: Extend validator and API calls to handle IPv6 addresses
- Persistent database: Store scan history in SQLite to track IP trends over time
- Geolocation map: Visualise IP origins on an interactive world map using Folium
- Automated alerting: Send email or Slack notifications on Malicious IP detection
- Scheduled scans: Automatic scans at configurable intervals using APScheduler
- SIEM integration: Export results compatible with Splunk, ELK Stack, and QRadar
- Firewall rule export: Generate ready-to-run iptables block rules from Malicious IPs
- User authentication: Login functionality for multi-analyst deployment
- REST API mode: Expose the scanner as an API callable by other security tools

Chapter 10: Conclusion

The Malicious IP Intelligence System successfully demonstrates how open threat intelligence APIs, modern Python web frameworks, and large language models can be combined into a practical, production-quality cybersecurity tool.

The system automates batch IP reputation checking, delivers instant visual risk classification through a professionally styled interactive dashboard, and provides AI-powered actionable mitigation guidance — capabilities directly applicable to real-world Security Operations Centre workflows.

All six identified robustness issues were resolved: the disconnected sidebar slider, missing input validation, zero-public-IP crash, low AI visibility, CSV overwriting, and single-status chart crash. The project demonstrates practical competency in REST API integration, Python data processing, Streamlit web development, applied AI, and cybersecurity domain knowledge.

> The modular architecture makes it straightforward to extend this system with additional intelligence sources, persistent storage, and enterprise integration features as described in Chapter 9.

Chapter 11: References

- AbuseIPDB API v2 Documentation. https://docs.abuseipdb.com/
- OpenAI API Documentation — Chat Completions. https://platform.openai.com/docs/
- Streamlit Documentation. https://docs.streamlit.io/
- Python requests Library. https://docs.python-requests.org/
- Pandas Documentation. https://pandas.pydata.org/docs/
- Matplotlib Documentation. https://matplotlib.org/stable/contents.html
- Python ipaddress Module. https://docs.python.org/3/library/ipaddress.html
- NIST Cybersecurity Framework v2.0. https://www.nist.gov/cyberframework
- OWASP Top 10 — 2021. https://owasp.org/Top10/
- Shodan Internet Intelligence Platform. https://www.shodan.io/
- Google Fonts — Share Tech Mono. https://fonts.google.com/specimen/Share+Tech+Mono
- Stix/TAXII Standards for Threat Intelligence Sharing. https://oasis-open.github.io/cti-documentation/
- git-hub -https://github.com/architcoder1234/Malicious-IP-Intelligence-System.git